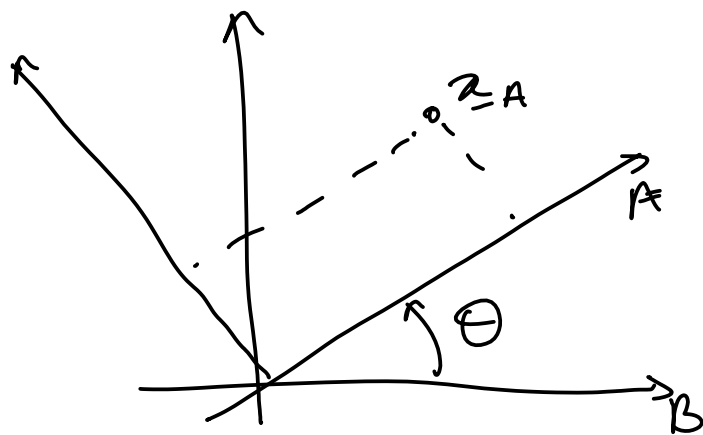
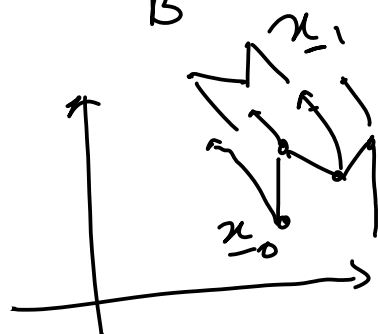




$$\underline{x}_B = R(\theta) \underline{x}_A$$

$$\underline{x}_1 = R(\theta) \underline{x}_0$$



2D Rotations

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

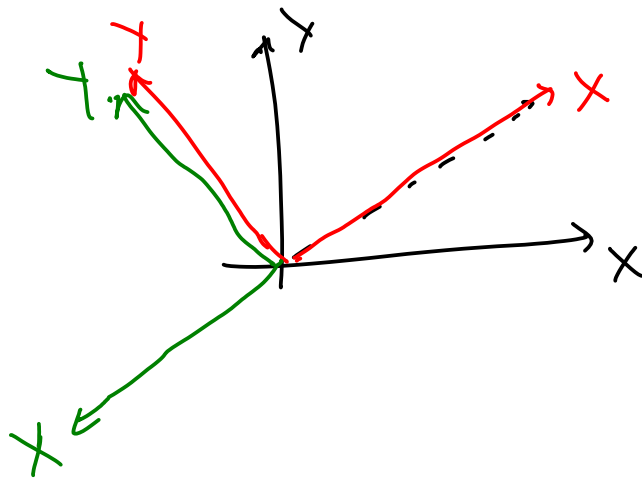
① Orthonormal Matrix

$$R^T R = I \quad R R^T = I$$

$$\det(O) = \pm 1$$

② $\det(R) = +1$

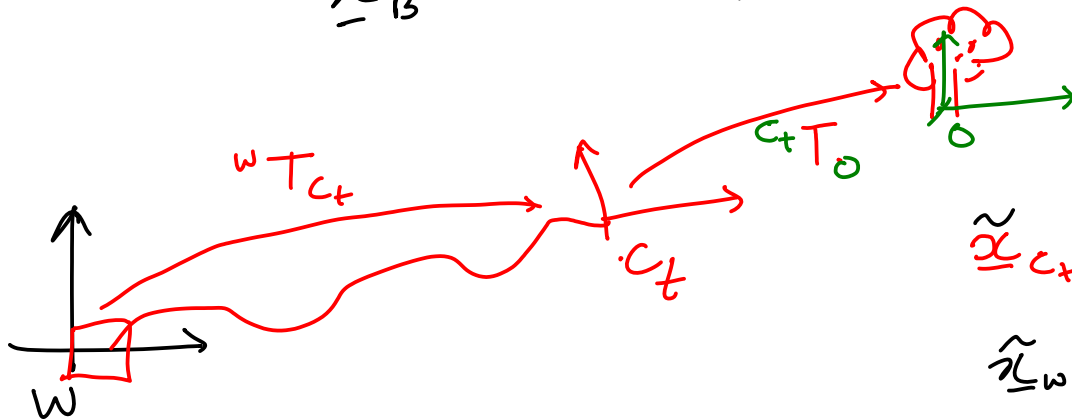
$$\det(\text{Reflection matrix}) = -1$$



2D Transformation

$$\begin{bmatrix} \underline{x}_B \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} {}^B R_A & {}^B t_A \\ \underline{0}^T & 1 \end{bmatrix}}_{\substack{\text{Rotation} \\ \text{translation}}} \begin{bmatrix} \underline{x}_A \\ 1 \end{bmatrix}$$

$$\tilde{\underline{x}}_B = {}^B T_A \tilde{\underline{x}}_A$$



$$\tilde{\underline{x}}_{C+} = {}^{C+}T_O \tilde{\underline{x}}_O$$

$$\tilde{\underline{x}}_W = {}^W T_{C+} \tilde{\underline{x}}_{C+}$$

$$\tilde{\underline{x}}_W = \underbrace{{}^W T_{C+} {}^{C+}T_O}_{{}^W T_O} \tilde{\underline{x}}_O$$

$${}^W T_O = {}^W T_{C+} {}^{C+}T_O \leftarrow \text{composition of transformation}$$

Given ${}^w T_{C_t}$ ${}^o T_{C_t}$
Find ${}^w T_o$?

$$\tilde{x}_o = {}^o T_{C_t} \tilde{x}_{C_t} \quad \tilde{x}_{C_t} = {}^{C_t} T_o \tilde{x}_o$$

$${}^o T_{C_t}^{-1} \tilde{x}_o = \tilde{x}_{C_t}$$

$$\boxed{{}^{C_t} T_o = {}^o T_{C_t}^{-1}}$$

Given $T = \begin{bmatrix} R & \underline{t} \\ \underline{0}^T & 1 \end{bmatrix}$, Find T^{-1} in terms of $R, \underline{t}$

$$\begin{bmatrix} \underline{y} \\ 1 \end{bmatrix} = \begin{bmatrix} R & \underline{t} \\ \underline{0}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{x} \\ 1 \end{bmatrix}$$

3x3

Solve for $\underline{x}$ in terms of $\underline{y}$, then write it as transformation matrix

$$\begin{bmatrix} \underline{y} \\ 1 \end{bmatrix} = \begin{bmatrix} R \underline{x} + \underline{t} \\ \underline{0}^T \underline{x} + 1 \end{bmatrix}$$

$$\underline{y} = R \underline{x} + \underline{t}$$

$$\underline{y} - \underline{t} = R \underline{x}$$

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

$$R^{-1} = R^T = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

$$\frac{\underline{y} - \underline{t}}{R} = \underline{x} \quad R^{-1}$$

Left multiply by R^{-1}

$$R^{-1}(\underline{y} - \underline{t}) = R^{-1}R\underline{x}$$

$$R^T(\underline{y} - \underline{t}) = \underline{x}$$

Find T^{-1}

$$R^T \underline{y} - R^T \underline{t} = \underline{x}$$

$$\left[\begin{array}{c|c} R^T & -R^T \underline{t} \\ \hline \underline{0}^T & 1 \end{array} \right] \begin{bmatrix} \underline{y} \\ 1 \end{bmatrix} = \begin{bmatrix} \underline{x} \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \underline{x} \\ 1 \end{bmatrix} = \begin{bmatrix} R^T & -R^T \underline{t} \\ \underline{0}^T & 1 \end{bmatrix} \begin{bmatrix} \underline{y} \\ 1 \end{bmatrix}$$

$$\hat{\underline{x}} = \underbrace{T^{-1}} \hat{\underline{y}}$$

Right multiply by R^{-1}

$$(\underline{y} - \underline{t}) R^{-1} = R \underline{x} R^{-1}$$

2×1 2×2 2×1 2×2

$$\underline{y} = R \underline{x} + \underline{t}$$

$$\underline{y} = \begin{bmatrix} R & \underline{t} \end{bmatrix} \begin{bmatrix} \underline{x} \\ 1 \end{bmatrix}$$

$$\hat{\underline{y}} = T \underline{x}$$

$$= \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix}$$

$$= R(-\theta)$$

3D Rotations

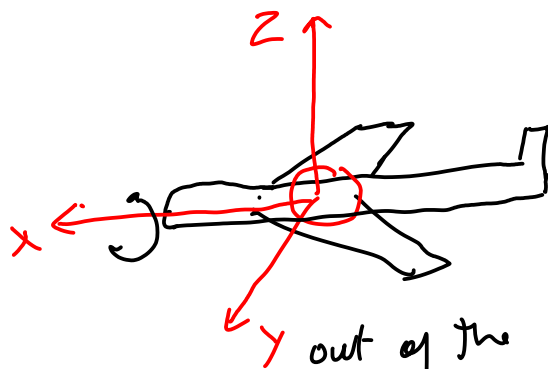
Euler Theorem

3D Rotat. ion = 3 x 2D Rotations

Roll = $R_x(\alpha)$

Pitch = $R_y(\beta)$

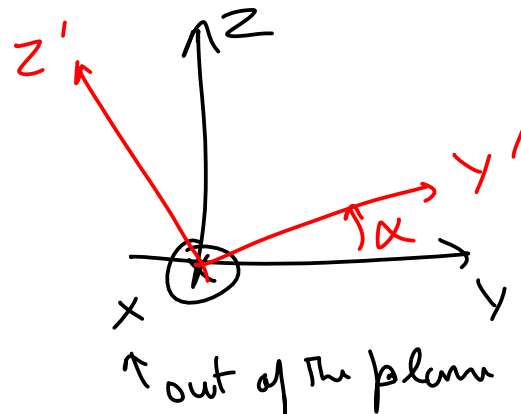
Yaw = $R_z(\gamma)$



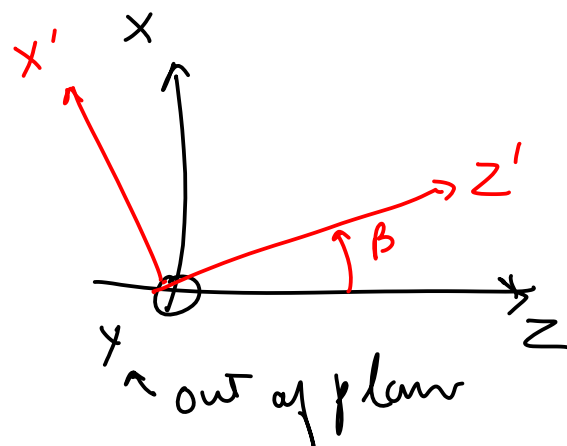
$$R_z(\gamma) = \begin{bmatrix} \cos(\gamma) & -\sin(\gamma) & 0 \\ \sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \underbrace{\begin{bmatrix} c\gamma & -s\gamma & 0 \\ s\gamma & c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R_z(\gamma)} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & -\sin(\alpha) \\ 0 & \sin(\alpha) & \cos(\alpha) \end{bmatrix}_{3 \times 3}$$



$$R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$



$$\underline{x} = R(\theta) \underline{x'}$$

$$\underline{x} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

